

Aggregate Employment Fluctuations with Microeconomic Asymmetries

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We provide a simple explanation for the observation from the U.S. manufacturing sector that the job destruction rate fluctuates more than the job creation rate. In our model, proportional plant-level costs of creating and destroying jobs cause shrinking plants to be more sensitive to aggregate shocks than growing plants. We describe circumstances in which this microeconomic asymmetry is preserved in the aggregate and show that it can account for much of the observed asymmetries in gross job flows. This is so even though we abstract from job matching frictions, incomplete contracts, and aggregate congestion effects. (JEL E00, L00, J00)

Consider the problem of choosing employment at a plant facing job creation and destruction costs which are proportional to the number of jobs created or destroyed, as in Samuel Ben-
tolila and Giuseppe Bertola (1990), Hugo Hopenhayn and Richard Rogerson (1993), and Bertola and Rogerson (1996). When there is a temporary change in the real wage, the plant's response depends on whether it is creating or destroying jobs. For an expanding plant, the total cost of the last job created is greater than the wage because of the expected adjustment costs incurred. These additional costs, which include the job creation cost and the expected cost of destroying that job in the future, lower the elasticity of the total cost of creating a job with respect to the wage below unity. For a

shrinking plant, the total cost of the last job retained is less than the wage because, by retaining the job, the plant avoids the cost of job destruction and the expected cost of creating that same job again in the future. The subtraction of avoided adjustment costs increases the elasticity of the total cost of retaining a job with respect to the wage above unity. Since expanding and contracting plants operate on the margins that equate the costs and benefits of adding and retaining a job, respectively, the asymmetric responses of the total costs of job creation and retention can induce contracting plants to respond more than expanding plants to wage changes.

We apply this insight to study the behavior of aggregate job creation and destruction rates, such as those measured by Steven J. Davis et al. (1996). Their observation that job destruction is more volatile than job creation in the U.S. manufacturing sector has challenged business-cycle theory, because standard models based on the one-sector stochastic growth model predict equally volatile job creation and destruction. To show how the addition of adjustment costs to such a model can reconcile business-cycle theory with the evidence, we construct a simple model economy with a fixed population of plants that face proportional costs of job creation and destruction.

Plants in the model produce a homogeneous good sold in a competitive goods market and hire workers in a competitive labor market. Technology shocks that are idiosyncratic to

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individual plants drive ongoing job creation and destruction. Otherwise, plants' technological opportunities are unchanged. In particular, plant obsolescence due to investment-specific technological change and the ensuing flow of workers from exiting to entering plants, as in Ricardo J. Caballero and Mohamad L. Hammour (1994, 1996) and Campbell (1998), play no role in the model. Shocks that disturb the real wage cause aggregate fluctuations in gross job flows. As described above, proportional adjustment costs induce employment decisions at growing plants to respond less to these shocks than those at shrinking plants.

The microeconomic asymmetry of the optimal employment policy does not necessarily translate into a similar asymmetry in gross job flows. This is because aggregation of heterogeneous plants also reflects the evolution of their distribution across the relevant state space and the properties of the aggregate disturbance. Caballero (1992) demonstrated this with a model similar to ours in which plants use a two-sided (S, s) policy. In his model, the endogenous evolution of the plant distribution exactly offsets the effects of an assumed microeconomic policy asymmetry on gross job flows.

To address the ambiguities inherent in aggregation, we simulate aggregate data from our model and compare it to empirical evidence on gross job flows. We show how the microeconomic policy asymmetry can be preserved in the aggregate so that the model reproduces asymmetries in gross job flows of the kind observed in the U.S. manufacturing sector. Caballero (1992) showed that if the process driving aggregate fluctuations is skewed so that recessions are relatively deep and infrequent, then his model can reproduce the excess volatility of job destruction. In our model the aggregate shock is symmetrically distributed, so these effects are not present. Christopher L. Foote (1998) has shown that another modification to the basic (S, s) model, adding a negative trend to each plant's desired employment, can also increase the variability of job destruction relative to job creation. In our model, trends influence job creation and destruction in the way suggested by Foote's results, but their effects are small relative to the impact of the microeconomic policy asymmetry.

The job creation and destruction costs that are

central to our analysis are easily motivated without appealing to contracting or job matching frictions, which are central components of the models of gross job flows analyzed by Dale T. Mortensen and Christopher A. Pissarides (1994) and Caballero and Hammour (1996). In our model, a plant pays no adjustment costs if the identities of its employees change but the number of employees remains constant. Therefore, the adjustment costs we study are net adjustment costs, as defined by Daniel S. Hamermesh and Gerard A. Pfann (1996). Following their interpretation, we view these costs as arising from the lost output incurred when reorganizing a plant's production process to operate at a larger or smaller scale. Hopenhayn and Rogerson (1993) and Bertola and Rogerson (1996) adopt a specification for job destruction costs that is similar to ours, but they interpret them as costs of firing workers. This interpretation requires that job destruction is identical to worker separations, as it is if the turnover of workers at a plant follows a last-in first-out policy.

As we show below, the optimal employment policy of a plant facing proportional net adjustment costs involves frequently leaving employment unchanged between periods. Hence, our adjustment cost specification is consistent with Hamermesh's (1989) observation that manufacturing plants adjust their employment only infrequently. Furthermore, using Dutch data, Hamermesh et al. (1994) found that many firms kept the total number of jobs constant over a two-year period but no firm kept the identity of its employees constant. Firms that did not change their employment replaced about 11 percent of their workforce over these two years.¹ These observations suggest the presence of net adjustment costs that do not depend on changing workers' identities. Finally, Edward P. Lazear (1995 p. 77) reports that "Human resource managers think in terms of slots or jobs, and think of these slots or jobs as being fundamental to the organization of the firm." This provides further, although indirect, evidence that the costs of creating and destroying jobs are significant.

¹ See the second row of Table 3 in Hamermesh et al. (1994).

The remainder of the paper proceeds as follows. The next section contains a simple example that illustrates how optimally chosen job creation and destruction can respond asymmetrically to aggregate shocks. Section II presents the model. In Section III we use calibrated versions of the model to investigate its implications for gross job flows. Section IV discusses the relationship between our work and previous studies of gross job flows. In Section V we conclude by discussing some implications of our findings for the study of business cycles and directions for future research.

I. An Illustrative Example

To build intuition for our findings, we study a simple example adapted from the steady-state model of Bertola and Rogerson (1996). It consists of a perfectly competitive industry populated by a continuum of plants that experience idiosyncratic productivity shocks and must pay job creation and destruction costs whenever they expand or contract employment. We demonstrate the importance of asymmetries in plant-level creation and destruction for gross job flows by studying the transitional dynamics between steady states induced by a permanent increase in the wage. This is equivalent to the case in which the wage is the numeraire and the price of the industry's output declines permanently. Due to the adjustment costs, the optimal employment policies of job-creating and job-destroying plants respond asymmetrically to the wage change along the transition path. This microeconomic asymmetry causes the industry to adjust employment primarily through job destruction.

A. Steady State

Consider a representative plant that produces a homogeneous good for sale in a competitive market. The plant hires labor, which comes in fixed shift lengths, in a competitive labor market. We refer to an employee working one shift as a job. The per-period cost of a job measured in units of the constant output price, which we call the wage, is denoted by W . We initially assume that W is constant. Let n_t denote the plant's employment at time t , and let z_t denote the plant's idiosyncratic productivity level. The

latter follows a Markov chain over the two states $z_g > z_b > 0$, the "good" and "bad" states, respectively. The probability of z_t changing states is constant and equals p . Output is given by $z_t n_t^\alpha$, where $\alpha \in (0, 1)$. The strict concavity of the production function reflects the presence of a fixed factor at the plant or a limit to a manager's span of control, as in Robert E. Lucas, Jr. (1978). When the plant changes the scale of its employment, it incurs adjustment costs. If it expands employment it incurs the job creation cost of τ_c units of lost output for every job added, and if it contracts employment it incurs the job destruction cost of τ_d units of lost output for every job subtracted. Finally, the plant manager is risk neutral, discounts future profits with the constant discount factor $\beta \in (0, 1)$, and maximizes the expected present discounted value of profits.

With these assumptions, it is straightforward to show that there exists an ergodic set for the plant's employment, which has exactly two points. To do so, suppose that $n_t = n_g$ whenever $z_t = z_g$ and $n_t = n_b$ whenever $z_t = z_b$. Then the expected present value of the marginal product of employment minus the wage follows a two-state Markov chain, taking on the values v_g and v_b in the good and bad productivity states, respectively. It follows that v_g and v_b must satisfy

$$(1) \quad v_g = \alpha z_g n_g^{\alpha-1} - W + \beta((1-p)v_g + p v_b)$$

$$(2) \quad v_b = \alpha z_b n_b^{\alpha-1} - W + \beta((1-p)v_b + p v_g).$$

If $n_g > n_b$, so that over time the plant both creates and destroys jobs, a necessary condition for the optimality of this employment policy is that $v_g = \tau_c$ and $v_b = -\tau_d$. That is, the marginal net revenue of creating or destroying a job must equal the associated marginal cost. Substituting these conditions into (1) and (2) yields

$$(3) \quad \alpha z_g n_g^{\alpha-1} = W + (1 - \beta(1-p))\tau_c + \beta p \tau_d$$

$$(4) \quad \alpha z_b n_b^{\alpha-1} \\ = W - (1 - \beta(1 - p))\tau_d - \beta p \tau_c.$$

These conditions have a simple interpretation. First consider (3). Suppose the plant's productivity switches from bad to good, and the plant considers increasing employment to n . The current net marginal benefit from this decision is $\alpha z_g n^{\alpha-1} - W - \tau_c$, the marginal product of the last job created less the wage and creation costs of this last job. The decision to create jobs in the current period also impacts future profits because of the adjustment costs. If productivity is good in the next period, the plant saves on the costs of creating the last job added in the current period. The discounted expected value of this saving is $\beta(1 - p)\tau_c$. If productivity is bad in the next period, the decision to add the last job in the current period involves higher job destruction than otherwise. The discounted expected value of this cost is $\beta p \tau_d$. Equation (3) implies that n_g is the level of employment that exactly equates the benefits and costs of adding an additional job when productivity is good. Similarly, (4) implies that n_b is the level of employment that exactly equates the benefits and costs of retaining an additional job when productivity is bad.

We now describe aggregate gross job flows in the steady state. Equations (3) and (4) can be solved for n_g and n_b , and it can be verified that both n_g and n_b are decreasing in W . We proceed under the assumption that n_g and n_b are well defined and that n_g is strictly greater than n_b . A plant destroys jobs whenever productivity changes from z_g to z_b , it creates jobs when the opposite change occurs, and it neither creates nor destroys jobs if its productivity stays the same. In the steady state, half of the plants are in the good state and the other half are in the bad state, with p plants changing productivity every period. For now we approximate the job creation and destruction rates defined by Davis and John C. Haltiwanger (1990) with the cross-sectional averages of $\ln(n_t/n_{t-1}) \times I\{n_t > n_{t-1}\}$ and $-\ln(n_t/n_{t-1}) \times I\{n_t < n_{t-1}\}$, respectively, where $I\{\cdot\}$ is an indicator function that equals one if its argument is true and equals zero otherwise. These approximations simplify the example considerably. We discuss

their adequacy below in Section II, subsection C. Using them, the steady-state aggregate rates of job creation, POS^* , and job destruction, NEG^* , both equal $(p/2)\ln(n_g/n_b)$.

B. Transitional Dynamics

To see how adjustment costs impact gross job flows, we consider the transition of the industry from an initial steady state to a new steady state with a slightly higher wage. Suppose that the wage increases unexpectedly and permanently from W to W' . One interpretation of this change is that it is the consequence of a change in workers' reservation wages. Another interpretation is that it is due to a change in demand for the industry's product. We provide a formal general-equilibrium interpretation of changes in W in Section II, subsection D.

To simplify the calculations, we assume that a plant must pay a small fixed cost whenever it changes its employment. This cost is small enough to ensure that the employment policy involving n_g and n_b is still optimal in the steady state, and it is large enough so that no plant wishes to change its employment after the wage change unless its productivity has also changed since then.² Given this auxiliary assumption, the employment policy following the wage change is as follows. Let n'_g and n'_b be solutions to (3) and (4) when W' replaces W . Employment is left at its level before the wage change until the plant experiences a productivity change. Thereafter, $n_t = n'_b$ whenever $z_t = z_b$, and $n_t = n'_g$ whenever $z_t = z_g$.

Because n_g and n_b are decreasing in W , we know that $n'_g < n_g$ and $n'_b < n_b$. To determine which employment level responds more, we

² To be precise, the fixed cost, $f > 0$, must satisfy the following conditions: (i) $v_i - f \geq \tilde{v}_i$, where $\tilde{v}_i$ is the value of delaying by one period the change in employment dictated by the $f = 0$ policy after the productivity state has just switched to i , $i = g, b$; (ii) $v'_i - f \geq \tilde{v}'_i$, where $\tilde{v}'_i$ denotes the value function associated with W' and $\tilde{v}'_i$ is defined analogously to $\tilde{v}_i$, $i = g, b$; (iii) $\hat{v}'_i \geq v'_i - f$, where $\hat{v}'_i$ denotes the value of delaying by one period the switch to n'_i from n_i after the wage change, $i = g, b$; (iv) $v'_i - f \geq \bar{v}'_i$, where $\bar{v}'_i$ denotes the value of delaying by one period the switch from n_j to n'_i after the wage change and after the productivity state has just switched from j to i , $j \neq i$ and $i, j = g, b$. For the numerical example that we study below to produce Figure 1, the set of f which satisfies all of these constraints is nonempty and equals $[0.0113, 0.0457]$.

compute the elasticities of n_b and n_g with respect to W using (3) and (4):

$$(5) \quad \partial \ln n_g / \partial \ln W = \frac{-1}{1 - \alpha} \frac{W}{W + (1 - \beta(1 - p))\tau_c + \beta p \tau_d}$$

$$(6) \quad \partial \ln n_b / \partial \ln W = \frac{-1}{1 - \alpha} \frac{W}{W - (1 - \beta(1 - p))\tau_d - \beta p \tau_c}.$$

Clearly, $\partial \ln n_b / \partial \ln W$ is greater in absolute value than $\partial \ln n_g / \partial \ln W$, so the policies of shrinking plants respond more than the policies of growing plants to the increase in the wage. We refer to this as policy asymmetry.

The asymmetric responses of n_g and n_b cause aggregate job creation and destruction to respond asymmetrically. In the period of the wage change, $p/2$ plants change employment from n_b to n'_g and $p/2$ plants change employment from n_g to n'_b . Hence, the job creation and destruction rates in the period of the wage change are

$$POS_0 = (p/2) \ln(n'_g/n_b)$$

$$NEG_0 = (p/2) \ln(n_g/n'_b).$$

Subtracting the common steady-state value of POS^* and NEG^* from both sides of these expressions yields

$$POS_0 - POS^* = (p/2) \ln(n'_g/n_g)$$

$$NEG_0 - NEG^* = (p/2) \ln(n_b/n'_b).$$

These expressions indicate that job creation falls and job destruction rises relative to their previous steady-state values. Because

$$|\partial \ln n_b / \partial \ln W| > |\partial \ln n_g / \partial \ln W|,$$

it follows that

$$|NEG_0 - NEG^*| > |POS_0 - POS^*|.$$

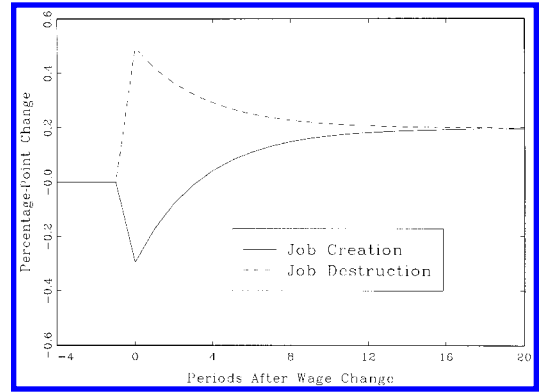


FIGURE 1. JOB CREATION AND DESTRUCTION ON THE TRANSITION FROM A LOW- TO A HIGH-WAGE STEADY STATE

That is, job destruction rises more than job creation falls following the increase in the wage. It is important to notice that this implication does not depend on the relative magnitude of the adjustment costs. In particular, it holds if one of the costs is zero while the other is strictly positive, and it holds for any strictly positive pair of values for the costs [see (5) and (6)].

The intuition for the asymmetric responses builds on the intuition underlying (3) and (4). The current and future adjustment costs associated with creating a job must be added to the wage to determine the total cost of creating a job, and this causes the total cost to respond less to changes in the wage than it would in the absence of adjustment costs. Hence a smaller change in n_g is required to maintain equality of the costs and benefits of creating a job, compared to the no-adjustment-cost case. In contrast, the current and future adjustment costs associated with destroying a job must be subtracted from the wage to determine the total cost of retaining a job. This means a larger change in n_b is required to maintain job destruction efficiency compared to the no-adjustment-cost case. Since n_g and n_b respond by the same amount in the no-adjustment-cost case, it follows that job destruction must respond by more than job creation in the presence of adjustment costs.

It is straightforward to show that the asymmetry persists beyond the date of the wage change. Figure 1 displays the deviation of the

job creation and destruction rates from their steady-state values along the transition path following a 1-percent increase in the wage under the assumption that $\beta = 1.05^{-1/4}$, $\alpha = 2/3$, $p = 1/4$, $\tau_c = \tau_d = 1/2$, $z_g = \exp(0.3)$, and $z_b = \exp(-0.3)$. In the period of the wage change, the job creation rate falls by 0.3 percentage points and the job destruction rate rises by 0.5 percentage points. Thereafter, job creation rises and job destruction descends toward their common, higher level in the new steady state.

This example demonstrates that the presence of proportional job creation and destruction costs can cause the aggregate job creation and destruction rates to respond asymmetrically to a permanent increase in the cost of a job. It is easy to verify that the same asymmetries arise with a permanent wage decrease. However, close inspection of the example reveals two interesting qualifications to the basic result. First, the policy asymmetry depends on assuming the adjustment costs are constant in units of output. This assumption corresponds to the notion of net adjustment costs described by Hamermesh and Pfann (1996), which we interpreted above as being the lost output incurred when reorganizing a plant's production process to operate at a larger or smaller scale. Hamermesh and Pfann (1996) also discuss gross adjustment costs such as training costs or any other cost that is incurred when the worker in a particular job changes. These costs can be modeled by assuming the adjustment costs are incurred in units of time. Under this assumption there would be no policy asymmetry since the adjustment costs would be proportional to the current wage and both of the elasticities in (5) and (6) would equal $-1/(1 - \alpha)$. Evidently, the presence of net adjustment costs is the ultimate source of the policy asymmetry.

The second qualification is that the magnitude of the policy asymmetry decreases with p , so that the responses of aggregate job creation and destruction become more similar as idiosyncratic shocks become more persistent. The intuition for this comes from inspection of the first-order conditions that define n_g and n_b , (3) and (4). For an expanding plant, one benefit of creating the marginal job is the future creation cost avoided if productivity remains good. This benefit partially offsets the current creation cost. One cost of creating the marginal job is the

destruction cost incurred if productivity becomes bad. This adds to the cost of job creation. When productivity is persistent, the offsetting benefit is very large and the additional cost is very small. Because the total adjustment costs incurred for the marginal job are smaller when p is small, n_g responds by more to changes in W . The analysis of the job destruction decision is similar. The adjustment costs avoided from retaining the marginal job are small when productivity is persistent, so n_b responds less to changes in W . Because the responses of n_g and n_b to a change in W become more similar when p is smaller, the responses of aggregate job creation and destruction also become more similar.

II. The Model

The example in the previous section illustrates how proportional job creation and destruction costs can give rise to asymmetries in gross job flows in response to aggregate shocks. To further assess this mechanism for understanding gross job flows, we modify the example in three ways. First, ongoing aggregate fluctuations are added. We assume that the wage, W_t , follows a Markov chain over the grid $W^h > W^l > 0$ with transition matrix **II**. Note that, as before, changes in the wage can reflect both "demand" and "supply" shocks. Second, we assume the distribution of idiosyncratic productivity shocks is continuous rather than discrete. Specifically, $\ln z_t$ follows a random walk with continuously distributed innovation, ε_t , which is independently and identically distributed (i.i.d.) across time and plants. We assume ε_t has a normal distribution truncated at very large and small values with mean μ and standard deviation σ_z . The assumption of a bounded support for ε_t is a technical condition used in the analysis of the plant's dynamic programming problem. Since plants' idiosyncratic productivity levels are very persistent, intuition from the previous section suggests that the model will be biased toward finding little asymmetry in gross job flows. We use the random-walk assumption because it simplifies the analysis considerably and also makes it easier to compare our model to conventional (S, s) models of employment dynamics. The third change to the example is that we assume that plants do not face fixed costs of changing employment.

While such costs simplified the analysis above, here they introduce unnecessary technical difficulties. In all other respects, the model described in this section is the same as the example.

The remainder of this section proceeds as follows. First, we describe the profit-maximization problem of the representative plant manager and the optimal employment policy it implies. After this we discuss the possible implications of the employment policy for aggregate gross job flow dynamics. Finally, we present a general-equilibrium version of the model, which we use to illustrate possible interpretations of fluctuations in the wage.

A. The Plant Manager's Problem

The problem faced by the representative plant manager is to choose employment at all dates to maximize the present discounted value of the plant's profits. To study this problem, we cast it as a dynamic program. Each period, the manager chooses current employment after observing current productivity, employment in the previous period, and the wage. Dropping time subscripts, let m and n denote the level of employment in the previous and current period, respectively, and define the scaled employment variables, $x = m/z^{1/(1-\alpha)}$ and $y = n/z^{1/(1-\alpha)}$. The dynamic program that summarizes the plant manager's profit-maximization problem is then

$$(7) \quad g(z, x, W) = \max_y z^{1/(1-\alpha)} \times [y^\alpha - Wy - \tau(y, x)(y - x)] + \beta E[g(z', y/u', W')|z, W].$$

Here we use the "'' notation in the usual fashion, $u' = (z'/z)^{1/(1-\alpha)} = \exp(\varepsilon'/(1-\alpha))$, and $E[\cdot | \cdot]$ denotes the conditional expectations operator. Finally, $\tau(y, x)$ represents the per-job adjustment costs, which, as before, are measured in units of the output good:

$$\tau(y, x) = \tau_c I\{y > x\} - \tau_d I\{y < x\}.$$

In Campbell and Fisher (1999) we show that

the value function $g(z, x, W)$ that corresponds to the solution to the sequence version of the manager's profit-maximization problem is homogeneous of degree $1/(1-\alpha)$ in z . Therefore, we can focus on solutions to (7) that satisfy this property and write $g(z, x, W) = z^{1/(1-\alpha)}v(x, W)$, where $v(x, W)$ is defined to equal $g(1, x, W)$. If we substitute this expression for $g(z, x, W)$ into (7) and use the fact that $\ln z$ follows a random walk, we can eliminate z from both sides of the equation. The resulting Bellman equation is

$$(8) \quad v(x, W) = \max_y y^\alpha - Wy - \tau(y, x)(y - x) + \beta E[u' v(y/u', W')|W].$$

The solution to (8) characterizes the solution to the plant manager's profit-maximization problem. After being adapted to this particular problem, standard dynamic programming arguments found in Nancy L. Stokey and Lucas (1989) can be used to demonstrate the existence and uniqueness of a function $v(x, W)$ that satisfies (8) and to show that this function is concave in x . This is discussed in more detail in Campbell and Fisher (1999).

B. The Optimal Employment Policy

We now describe the policy function for y and show how it is used to formulate the manager's optimal employment policy. Because $v(x, W)$ is concave in x (and so differentiable in x almost everywhere) and u has a continuous distribution, $E[u' v(y/u', W')|W]$ is a differentiable function of y .³ The differentiability of the expected value function implies that the objective on the right-hand side of (8) is differentiable in y everywhere except at $y = x$. Therefore, the optimal choice of y must satisfy the first-order conditions,

³ This is proven in Campbell and Fisher (1999). Note that the arguments of Lawrence M. Benveniste and José A. Scheinkman (1979), which rely on the differentiability of the return function to establish differentiability of the value function, do not hold here.

$$(9) \quad \alpha y^{\alpha-1} - W - \tau_c + \beta E[v_1(y/u', W')|W] \leq 0,$$

with equality if $y > x$, and

$$(10) \quad \alpha y^{\alpha-1} - W + \tau_d + \beta E[v_1(y/u', W')|W] \geq 0,$$

with equality if $y < x$. Here, $v_1(x, W)$ is defined as one of the value function's directional derivatives, which exist everywhere due to its concavity.

Define the *scaled employment thresholds* $\underline{y}(W)$ and $\bar{y}(W)$ as the values of y that satisfy the derivative conditions in (9) and (10) with equality.⁴ Notice that $\underline{y}(W) < \bar{y}(W)$ if either τ_c or τ_d is positive. These functions characterize the optimal scaled employment choice associated with (8), y^* . If $\underline{y}(W) \leq x \leq \bar{y}(W)$, then employment is unchanged, $y^* = x$. If $x < \underline{y}(W)$, then $y^* = \underline{y}(W)$ and if $x > \bar{y}(W)$, then $y^* = \bar{y}(W)$. With this policy and the definitions of x and y in hand, it is straightforward to construct the optimal policy for the unscaled level of employment, n^* . This can be described in terms of the *job creation schedule*,

$$\underline{n}(z, W) = \underline{y}(W)z^{1/(1-\alpha)}$$

and the *job destruction schedule*,

$$\bar{n}(z, W) = \bar{y}(W)z^{1/(1-\alpha)}.$$

Figure 2 illustrates the optimal employment policy for a given value of W . The job creation and destruction schedules are both loglinear in z with intercepts $\ln \underline{y}(W)$ and $\ln \bar{y}(W)$ and common slope $1/(1 - \alpha)$. Since $\ln \underline{y}(W) < \ln \bar{y}(W)$, the job creation schedule lies below the job destruction schedule. Consider the employment decisions at three plants with identical lagged employment but different realizations of technology in the current period. These plants are denoted A, B, and C in the figure. Plant A lies above the job destruction schedule. It chooses current employment to be the

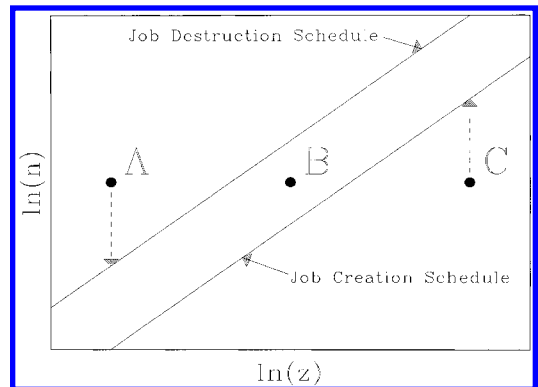


FIGURE 2. OPTIMAL EMPLOYMENT POLICY

value implied by the destruction schedule at its current level of technology. Therefore, it destroys jobs at the logarithmic rate equal to the vertical distance from A to the job destruction schedule. Plant B lies between the two schedules, so it leaves employment unchanged. Plant C lies below the job creation schedule. It creates jobs at the logarithmic rate equal to the vertical distance from C to the job creation schedule.

C. Gross Job Flow Dynamics

We are interested in the dynamics of gross job flows implied by a continuum of plants following the optimal employment policy just described. Davis and Haltiwanger (1990) define the two gross job flow rates as the sum of all jobs created (or destroyed) divided by the average of aggregate employment in the current and previous period. In our model these rates are

$$(11) \quad POS_t = \frac{2}{N_t + N_{t-1}} \times \int_0^1 (n_t(i) - n_{t-1}(i))^+ di,$$

$$(12) \quad NEG_t = \frac{2}{N_t + N_{t-1}} \times \int_0^1 |(n_t(i) - n_{t-1}(i))^-| di.$$

⁴ Because $\alpha y^{\alpha-1}$ and $E[v_1(y/u, W')|W]$ are decreasing functions of y , both $\underline{y}(W)$ and $\bar{y}(W)$ are well defined. We exploit the envelope condition to formulate algebraic expressions for $\underline{y}(W)$ and $\bar{y}(W)$, which are straightforward to compute numerically. See Campbell and Fisher (1999) for details.

Here we give each plant in the industry a name on the unit interval and use $n_t(i)$ to denote plant i 's employment at date t . Also, a superscript of "+" or "-" on any random variable denotes its positive or negative parts, respectively. Finally, $N_t = \int_0^1 n_t(i) di$ denotes date t aggregate employment.

Since the wage determines $\ln \bar{y}(W)$ and $\ln \underline{y}(W)$, it follows that variation in the wage leads to fluctuations in the creation and destruction schedules. The example in Section I suggests that $\ln \underline{y}(W)$ will respond by less to a wage change than $\ln \bar{y}(W)$, so we expect the creation schedule to be less variable than the destruction schedule. That is, we expect there to be a policy asymmetry so that the employment decisions of job-creating plants are less volatile than the employment decisions of job-destroying plants.

Figure 3 is helpful for understanding how policy asymmetry can lead to asymmetric volatility of aggregate job creation and destruction. It contains hypothetical creation and destruction schedules for the two wage states. The top solid line is $\ln \bar{n}(z, W^l)$ and the bottom solid line is $\ln \underline{n}(z, W^l)$. Similarly, the top and bottom dashed lines are $\ln \bar{n}(z, W^h)$ and $\ln \underline{n}(z, W^h)$, respectively. The dashed arrows indicate the movement of the schedules when the wage changes from W^l to W^h . As they are drawn, the destruction schedule responds by more than the creation schedule to a wage change.

Consider the employment decisions of four plants with employment and technology as indicated by the labeled points in Figure 3. Plant A destroys jobs regardless of whether the wage changes, but when the wage changes it destroys more jobs than otherwise. In the absence of a wage change, plant B keeps its employment constant, but it destroys jobs when the wage changes. Plants C and D both create jobs in the absence of the wage change. When the wage changes, plant C leaves its employment unchanged and plant D creates fewer jobs than otherwise. Because the destruction schedule shifts by more than the creation schedule, the intensity of job destruction is increased at plants that behave like A more than the intensity of job creation is reduced at plants like D. Furthermore, it is possible that the number of plants like B, which switch from inactivity to job destruction when the wage changes, is greater than the number of plants like C, which switch

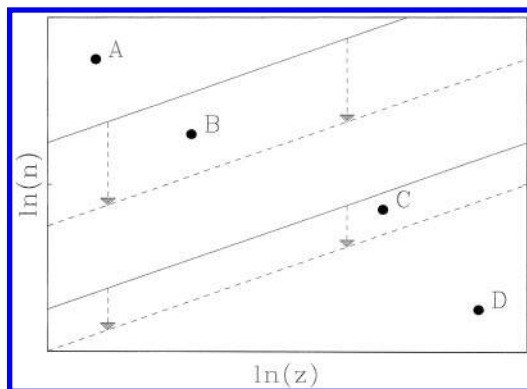


FIGURE 3. RESPONSE OF EMPLOYMENT POLICY TO AN INCREASE IN THE WAGE

from job creation to inactivity. These considerations suggest that job destruction will be more variable than job creation in the aggregate.

The intuition from Figure 3 is helpful but incomplete, because it describes changes in logarithmic growth rates, $\ln(n_t(i)/n_{t-1}(i))$, rather than actual growth rates,

$$g_t(i) = n_t(i)/n_{t-1}(i) - 1.$$

By dividing and multiplying the integrands in (11) and (12) by $n_{t-1}(i)/N_{t-1}$, POS_t and NEG_t can be written as size-weighted averages of the positive and negative parts of plants' actual growth rates:

$$(13) \quad POS_t$$

$$= \frac{2N_{t-1}}{N_t + N_{t-1}} \int_0^1 \frac{n_{t-1}(i)}{N_{t-1}} g_t(i)^+ di,$$

$$(14) \quad NEG_t$$

$$= \frac{2N_{t-1}}{N_t + N_{t-1}} \int_0^1 \frac{n_{t-1}(i)}{N_{t-1}} |g_t(i)^-| di.$$

While the growth rates of macroeconomic time series are usually close enough to zero for a logarithmic approximation to be valid, the growth rates of microeconomic agents are often far from zero, both in our model and in the data. This suggests it is important to use the

actual-growth-rate expression when computing aggregate gross job flows.

To appreciate the consequences of using actual growth rates rather than logarithmic growth rates in the calculation of gross job flows, consider the responses of plants A and D from Figure 3. Their logarithmic growth rates are

$$\begin{aligned} \ln(n_t(A)/n_{t-1}(A)) \\ &= \ln \bar{y}_t - \ln y_{t-1}(A) + \ln u_t(A), \\ \ln(n_t(D)/n_{t-1}(D)) \\ &= \ln \underline{y}_t - \ln y_{t-1}(D) + \ln u_t(D). \end{aligned}$$

Note that here we replace explicit dependence of variables on the wage with date subscripts. The derivatives of the two logarithmic growth rates with respect to $\ln \underline{y}_t$ and $\ln \bar{y}_t$ both equal one. Hence, any asymmetry in how the intercepts of the policy schedules respond to a shock will translate directly to the response of growing and shrinking plants' logarithmic growth rates.

The actual growth rates of the two plants are

$$\begin{aligned} g_t(A) \\ &= \exp(\ln \bar{y}_t - \ln y_{t-1}(A) + \ln u_t(A)) - 1, \\ g_t(D) \\ &= \exp(\ln \underline{y}_t - \ln y_{t-1}(D) + \ln u_t(D)) - 1. \end{aligned}$$

Since $\exp(\cdot)$ is a convex function and plant A 's growth rate is less than plant D 's, plant D 's actual growth rate is more responsive to a change in $\ln \underline{y}_t$ than is plant A 's actual growth rate to a change in $\ln \bar{y}_t$, i.e.,

$$\frac{\partial g_t(A)}{\partial \ln \bar{y}_t} = g_t(A) + 1 < g_t(D) + 1 = \frac{\partial g_t(D)}{\partial \ln \underline{y}_t}.$$

This means that if the intercepts of the job creation and destruction schedules respond to a shock by equal amounts, shrinking and growing plants' actual growth rates will respond asymmetrically, with growing plants exhibiting more sensitivity to the shock. If all else were held equal, this growth-rate asymmetry would tend

to decrease the variability of job destruction relative to job creation.⁵

The logarithmic approximations of aggregate job creation and destruction that we used in Section I do not incorporate this growth-rate asymmetry, so they tend to emphasize asymmetries in the employment policy. This was useful for illustrative purposes but is clearly inadequate for a quantitative evaluation of the model. Consequently, below we use (13) and (14) to compute gross job flows. We find that in some versions of the model the policy asymmetry is present, but relatively small. In these cases, the growth-rate asymmetry dominates our calculations, so that job creation is actually more volatile than job destruction. This suggests that policy asymmetry due to proportional adjustment costs fights an uphill battle to reproduce the observed relative volatility of job creation and destruction in manufacturing industries.

D. General Equilibrium

The industry-equilibrium model can be reinterpreted in a general-equilibrium framework with particular assumptions on tastes and technology. In this subsection we present such a framework. We use it to illustrate possible sources for the shocks driving job flows in the industry-equilibrium model and to show that the job flows in this model can be interpreted as those for a particular industry or as those for the economy as a whole.

⁵ The intuition just described addresses the size-weighted averages in (13) and (14), and neglects the common term in front of these expressions. This leading term equals two divided by the date t aggregate (actual) gross employment growth rate plus one. Since the aggregate growth rate is positively correlated with POS_t and negatively correlated with NEG_t , movements in it offset variation in the size-weighted average of $g_t(i)^+$, lowering the variance of POS_t , and augment variation in the size-weighted average of $|g_t(i)^-|$, increasing the variance of NEG_t . In the special case with no adjustment costs and no trend in aggregate employment, one can show that the asymmetric effects of this term on POS_t and NEG_t exactly offset the growth-rate asymmetry so that POS_t and NEG_t respond equally to wage shocks around the nonstochastic steady state. Increasing adjustment costs reduces both the growth-rate asymmetry and the volatility of the aggregate growth rate. However, the impact on the latter is stronger so that increasing adjustment costs diminishes the importance of the leading term for gross job flows relative to the growth-rate asymmetry.

The general-equilibrium economy is populated by a fixed continuum of households and a fixed continuum of plants. The households are identical and value sequences of a composite consumption good, $c_t > 0$, and work, $l_t \geq 0$, according to

$$E_0 \left[\sum_{t=0}^{\infty} \beta^t (c_t - A_t l_t) \right].$$

Here, the marginal disutility of work, A_t , is always positive and E_0 is the conditional expectation operator. In accordance with our assumption of fixed shift lengths made earlier, we restrict l_t to equal either 0 or 1 for any given household. The composite consumption good is derived as a linear combination of a finite number of nonstorable goods. For simplicity we suppose that there are just two of these goods, $c_{1,t}$ and $c_{2,t}$. The consumption good index is given by

$$c_t = a_{1,t} c_{1,t} + a_{2,t} c_{2,t},$$

$$a_{1,t} = 1 - a_{2,t}, \quad 0 < a_{2,t} < 1.$$

Plants, which are owned by households, are capable of producing either $c_{1,t}$ or $c_{2,t}$. Otherwise they are identical and consist of the technology described in Section II. The exogenous shocks $\{a_{i,t} : i = 1, 2\}$ and A_t evolve as Markov chains. Households and plant managers trade in competitive goods and labor markets and act accordingly. We assume a complete set of markets in state-contingent claims. To conclude the description of the model, a competitive equilibrium can be defined in the usual way.

In a competitive equilibrium in which there are always some consumers enjoying leisure and both sectors always have positive employment, preferences determine prices independently of the distribution of employment and idiosyncratic productivity across plants. Given that the plants are owned by households, plant managers discount future profits with the gross interest rate $1/\beta$. The wage denominated in units of the composite good is $w_t = A_t$, and the similarly denominated consumption good

prices are $p_{1,t} = 1 - a_{2,t}$ and $p_{2,t} = a_{2,t}$. The product wage for plants in industry one is $w_{1,t} = w_t/p_{1,t} = A_t/(1 - a_{2,t})$ and for plants in industry two, $w_{2,t} = A_t/a_{2,t}$.

Assume the plants in the industry-equilibrium model produce, say, good two. Then, by setting $W_t = w_{2,t}$ in that model, the implied gross job flows correspond to those in industry two in the general-equilibrium model. According to this interpretation, W_t reflects the labor-supply shock, A_t , and the consumption relative preference shock, $a_{2,t}$. Thus, the wage in the industry-equilibrium model, W_t , can be interpreted as a function of both "demand" and "supply" shocks. If $a_{2,t}$ is a constant, then gross job flows in the industry-equilibrium model correspond to the industry-specific and economywide job flows derived from the general-equilibrium model. Therefore, gross job flows in the industry-equilibrium model can be interpreted as coming from a particular industry or from the economy as a whole. Note that in the former case the consumption wage in the general-equilibrium model is not necessarily perfectly correlated with the object called the wage in the industry-equilibrium model.

III. Implications of the Model for Gross Job Flows

We have argued that the model developed in the previous section has the potential to generate asymmetries in gross job flows. Now we assess this potential by studying the gross job flows implied by various parameterizations of the model. Our main finding is that the model can generate asymmetries in gross job flows of the type observed in the U.S. manufacturing sector. These asymmetries arise due to quantitatively significant asymmetric variation in the creation and destruction schedules, which is strongest when wage changes are not expected to persist for very long.

This section is divided into three parts. In the first, we describe how we calibrate the model. After this we study several experiments designed to establish the model's basic qualitative implications. Finally, we analyze statistics on gross job flows implied by simulating several calibrated versions of the model. To judge the quantitative impact of adjustment costs, we compare the simulated data to quarterly data on

gross job flows for the transportation equipment industry (SIC 37). We compare the model to industry-level data since we prefer to view it as characterizing a single industry, rather than the entire economy, as is also consistent with the general-equilibrium framework just described. We confine our analysis to the transportation equipment industry, which is manufacturing's largest 2-digit industry, to conserve space. The results for other manufacturing industries for which data on gross job flows are available are qualitatively similar.

A. Parameter Values

To implement our model we need to specify the following parameters:

Plant-level parameters: α , β , μ , σ_z , τ_c , τ_d ;

Aggregate uncertainty: Π , W^h , W^l .

The elasticity of production with respect to labor input, α , is approximately equal to the share of output paid to labor.⁶ We set α to 0.66. The discount factor, β , is set to $1.05^{-1/4}$ so that the annual real interest rate is 5 percent and a period in the model corresponds to one quarter of a year.

The mean innovation to technology, μ , is chosen so that the model implies the same average net employment growth as in the transportation equipment industry for our sample, -0.11 percent.⁷ Holding fixed all other model parameters, the job reallocation rate in the model is increasing in the amount of idiosyncratic uncertainty. Therefore, we choose the innovation standard deviation, σ_z , by matching the model to the average job reallocation rate (the sum of the rates of job creation and destruction) for the transportation equipment industry, 11.3 percent.

We consider several specifications of the pro-

cess governing the wage. In each case Π has the basic structure

$$(15) \quad \Pi = \begin{bmatrix} \frac{1+\rho}{2} & \frac{1-\rho}{2} \\ \frac{1-\rho}{2} & \frac{1+\rho}{2} \end{bmatrix},$$

and the two wage states are expressed as $W^h = \exp(\sigma_w)$ and $W^l = \exp(-\sigma_w)$. Here ρ and σ_w are the first-order autocorrelation coefficient and the standard deviation of $\ln(W_t)$, respectively. Note that this specification implies that the wage is symmetrically distributed, unlike Caballero's (1992) asymmetric driving process. In our baseline calibration we select σ_w and ρ to match the standard deviation of employment growth and the autocorrelation of employment growth for the transportation equipment industry (the magnitudes are 3.55 percent and 0.08). In alternative calibrations where ρ is fixed a priori we continue to choose σ_w to match the standard deviation of employment growth.

Finally, consider the adjustment cost parameters, τ_c and τ_d . Recall that τ_c and τ_d are costs of changing the number of employees at a plant. These net adjustment costs involve disruptions to production and all other costs that are not related to the identity of a worker but depend solely on changing the number of employees. As Hamermesh and Pfann (1996) emphasize, net adjustment costs are intrinsically difficult to measure because usually they are implicit, in that they result in lost output, and thus are not measured and reported by firms. This complicates the task of assigning plausible values to τ_c and τ_d . Our first step in calibrating these parameters is to review briefly the available empirical evidence to determine plausible upper and lower bounds. We choose our baseline specification to be within these bounds. To assess the robustness of the results and intuition derived from the baseline calibration, we vary τ_c and τ_d between the chosen upper and lower bounds and recalibrate the model's other parameters.

There are three main sources of information on adjustment costs: anecdotal evidence, survey research, and formal econometric evidence. C. Lanier Benkard's (1998) study of the Lockheed L-1011's production provides one anecdote

⁶ This will not be exact because of the adjustment costs.

⁷ Without aggregate uncertainty, aggregate employment growth is given by the mean of u , $\exp(\mu/(1-\alpha) + \sigma_z^2/(2(1-\alpha)^2))$. With aggregate uncertainty, average employment growth may differ from this formula, but given σ_z and the other model parameters, μ still influences its magnitude as the formula suggests.

from the transportation equipment industry. He describes the organizational impact of Lockheed's contract with its union, which specified strict seniority privileges in hiring. To honor these privileges, the creation or destruction of a job that was not an entry-level position required extensive reassignment of incumbent workers among existing jobs. Such reassignment could involve up to ten positions to create one job.⁸ Ana M. Aizcorbe (1992) contends that the reassignment of workers across tasks in car assembly plants is an integral part of their line speed changes. The loss of output as reassigned workers learn their new jobs is one leading candidate source of net adjustment costs.

These anecdotes suggest that net adjustment costs can be large in the transportation equipment industry, but they provide nothing regarding the costs' precise magnitudes. Unfortunately, the existing literature that characterizes the magnitude of adjustment costs provides only a little additional information. Most studies based on firm- or plant-level observations use more readily available data on gross adjustment costs, those costs incurred when hiring or dismissing an individual worker. If workers separate from employers according to a last-in first-out rule, then the distinction between net and gross adjustment costs in the plant's profit-maximization problem disappears. This suggests that it is helpful to review what we do know from the existing literature, even though the correspondence between net and gross adjustment costs is far from exact.

Consider first the survey evidence. Hamermesh (1993) reports the results from several surveys of hiring and dismissal costs for employers in the United States. One survey of Los Angeles employers in 1980 implies that hiring and training costs were 0.91 and 1.61 of production and salaried workers' average quarterly earnings in 1980, respectively. In contrast, firing costs as fractions of quarterly earnings were only 0.07 and 0.21 for production and salaried workers, respectively.⁹ Another survey of firms' sever-

ance payments found that 87 percent of manufacturing firms paid severance payments to employees and that the median benefit per employee was 0.49 when expressed as a fraction of manufacturing production workers' quarterly earnings.¹⁰

Now consider the estimates based on econometric models. Estimates based on representative producer models with quadratic adjustment costs, such as Thomas J. Sargent (1978) and Matthew D. Shapiro (1986), are not directly informative regarding our model in which heterogeneous plants face proportional adjustment costs. Pfann and Bart Verspagen (1989) estimate an asymmetric but smooth adjustment cost function using data on Dutch firms' net employment changes and their accounting charges for restructuring. For their sample, they find that job creation costs are larger than job destruction costs, as in our baseline specification described below. Using Spanish data on employment of permanent and temporary workers, Victor Aguirregabiria and Cesar Alonso-Borrego (1998) estimate a model very similar to ours, but with stationary idiosyncratic productivity shocks and gross rather than net adjustment costs. They estimate the costs of hiring and firing permanent workers divided by their average quarterly earnings to equal 0.93 and 1.96. Their estimate of the hiring cost is surprisingly similar to that obtained from the Los Angeles survey for production workers, while that of the firing cost is unsurprisingly much larger. Lazear (1990) uses nonstructural econometric tools to investigate the impact of various job security provisions for several industrial

ism Manual of the Los Angeles Merchants and Manufacturers Association. Unfortunately, this primary source is unavailable to us. Hamermesh reports these costs in 1990 dollars. We converted these into 1980 dollars using the GDP implicit price deflator. To calculate these costs as fractions of average quarterly manufacturing earnings in 1980, we used employment and payroll data for manufacturing production and nonproduction workers from the *1980 Annual Survey of Manufactures* as reported in Table 1-1a of U.S. Bureau of the Census (1996).

⁸ See Benkard (1998 p. 7).

⁹ The survey reports average hiring and training costs for production workers of \$3,231 and salaried workers of \$8,720. Average firing costs were \$234 for production workers and \$1,125 for salaried workers. Hamermesh (1993) cites these figures from the *Turnover and Absentee-*

¹⁰ The median benefit per employee in this survey, Bureau of National Affairs, Inc. (1986) is \$2,486. To express the estimated benefit as a fraction of average quarterly earnings, we used data on manufacturing employees and wages from the *1986 Annual Survey of Manufactures*, as reported in Table 1-1a of U.S. Bureau of the Census (1996).

countries over the period 1956 to 1984. This work gives an indication of how large firing costs can be. For example, Lazear (1990 Table 1) finds that the average mandatory severance pay for a worker with ten years of tenure, measured in months' wages, was 1.66 in Germany, 5.24 in France, 8.24 in Israel, and 13.56 in Spain.

The well-known low dismissal costs faced by employers in the United States and the low firing-cost estimates from the Los Angeles survey motivate setting a lower bound of zero for τ_d . Because the available estimates of hiring costs are typically greater than firing costs in the United States, and hiring costs differ across industries, we set our lower bound for τ_c to 0.5, below the common Los Angeles/Aguirregabiria and Alonso-Borego estimate and above zero. We bound both τ_c and τ_d from above by three quarters' average wage payments, which is in line with Lazear's (1990) estimates of mandatory dismissal costs. This upper bound falls between the two values of the dismissal costs used by Hopenhayn and Rogerson (1993): two and four quarters' average wages. Our baseline specification takes into account these bounds and the observation that costs of job creation are higher than those of job destruction in the United States. We set τ_c and τ_d equal to 1.5 and 0.5, respectively, in our baseline calibration. Combined with a calibrated job reallocation rate of 11.3 percent, these choices imply that adjustment costs incurred by the industry average about 11.3 percent of the quarterly wage bill.¹¹

The parameters underlying our baseline calibration and a benchmark case without adjustment costs are summarized in Table 1. The latter parameterization is based on calibrating the model as described above assuming $\tau_c = \tau_d = 0$. When adjustment costs are zero, employment decisions at plants are governed by the usual static labor-demand schedule. Since it is not possible to match the persistence in job growth for any value of ρ in this case, we set ρ to the value implied by the baseline calibration. Notice the impact of adjustment costs on the values of the calibrated parameters. First, σ_z is more than three times larger in the baseline case

TABLE 1—PARAMETER VALUES

Parameter	Costly adjustment	Frictionless adjustment
τ_c	1.5	0
τ_d	0.5	0
α	0.66	0.66
μ_z	-0.0385	-0.00358
σ_z	0.161	0.0467
β	$1.05^{-1/4}$	$1.05^{-1/4}$
σ_w	0.0855	0.0117
ρ	0.466	0.466

compared to the frictionless case. This is due to the fact that, other things equal, fewer plants adjust employment in any given period with adjustment costs compared to the frictionless case. The larger value of σ_z is required to offset this effect. Second, σ_w is much larger in the baseline case as well. This is due to the fact that, other things equal, the presence of adjustment costs dampens plants' responses to a given wage change.

B. Illustrating the Basic Mechanism

Here we study the dynamics of gross job flows implied by the optimal employment policy for two polar cases for the wage transition matrix. The first case we consider involves specifying $\rho = -1$ so that the off-diagonal elements of Π equal unity. This implies that changes in the wage are known with certainty to last only one period. That is, the wage follows a two-period cycle, which roughly corresponds to the deterministic cycle studied in Caballero and Hammour (1996). The other parameters were calibrated as described above.¹² To assess the impact of adjustment costs, we also examine a frictionless version of this case in which all parameters are left at their calibrated values except for the adjustment costs, which are set to zero.

Simulated gross job flows implied by the two parameterizations are illustrated in Figure 4, where the left plot corresponds to the frictionless parameterization and the right plot corresponds to the model with positive adjustment

¹¹ We thank Fabio Schiantarelli for pointing out this implication of adjustment costs in our model.

¹² Values for the parameters not given in the text are $\mu = -0.0389$ percent, $\sigma_z = 16.2$ percent, and $\sigma_w = 10.4$ percent.

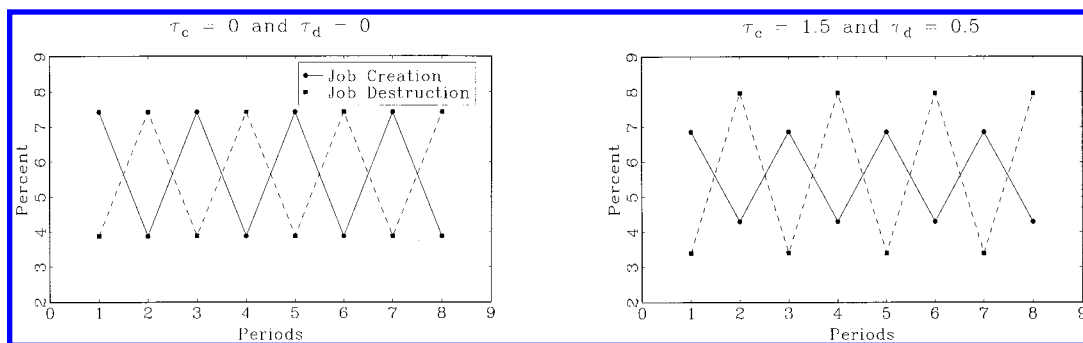


FIGURE 4. GROSS JOB FLOWS IN THE TWO-PERIOD-CYCLE MODEL

costs. This figure shows that, regardless of whether there are adjustment costs or not, job creation and destruction move in opposite directions every period. Job destruction is highest when the wage equals W^h , and job creation is highest when the wage equals W^l . Without adjustment costs the absolute value of the changes in job creation and job destruction are essentially the same every period so that these variables are roughly equally volatile. The adjustment costs act to dampen fluctuations in both job creation and job destruction. Moreover, job creation is dampened by more than job destruction so that job destruction is more variable than job creation when there are adjustment costs.

The symmetry of gross job flows in the frictionless model can be traced directly to the fact that in this model the job creation and destruction schedules are identical. Positive adjustment costs break the tight connection between the creation and destruction decisions in the frictionless model and introduce a significant asymmetry into how they respond to wage changes. In particular, the difference between the intercepts of the two destruction schedules, $\ln \bar{y}(W^l) - \ln \bar{y}(W^h)$, is 0.22, while the difference between the intercepts of the two creation schedules, $\ln \underline{y}(W^l) - \ln \underline{y}(W^h)$, is 0.10; the response of the destruction schedule to a wage change is more than double the corresponding response of the creation schedule. This policy asymmetry induces the intensity of job destruction at shrinking plants to change by more than the intensity of job creation at growing plants when the wage changes. The growth-rate asymmetry is also operational here and this partially

offsets the policy asymmetry so that, while job destruction is more volatile than job creation, it is less than twice as volatile.

The polar opposite of the two-period cycle for the wage is to suppose that when computing their optimal employment policy, plant managers expect the current wage to prevail forever. This is captured by assuming $\rho = 1$ in (15). We now examine how gross job flows respond to a once and for all unanticipated change in the wage, holding fixed the other parameters at the values used in the two-period cycle examples. This experiment is analogous to the example discussed in Section I, subsection B. Unlike in that example, the growth-rate asymmetry is operational here and offsets the policy asymmetry. Figure 5 plots job creation and destruction for changes in the wage from W^l to W^h (solid line) and from W^h to W^l (dashed line) for the case with no adjustment costs (left column) and the case with positive adjustment costs (right column). In each case the change in the wage occurs after first simulating the model for several periods at the initial wage until employment and the distribution of employment across plants converge. Gross job flows are plotted relative to their values before the change.

Consider the frictionless cases first. As in the two-period cycle examples, wage changes induce movements in creation and destruction in opposite directions. Job destruction rises and job creation falls following a wage increase, and they reverse their roles following a wage decrease. After one period, job creation and destruction return to their before-wage-change values. In both cases, the magnitude of the initial change in the gross job flow that

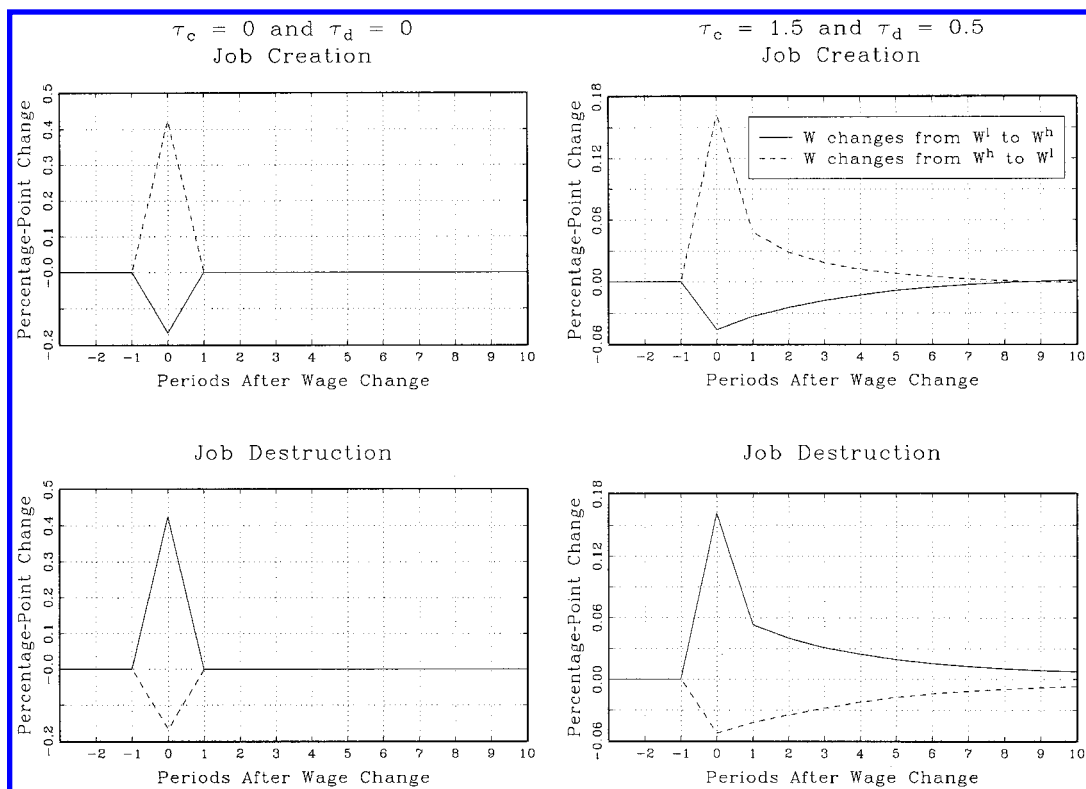


FIGURE 5. DYNAMICS OF GROSS JOB FLOWS IN RESPONSE TO PERMANENT WAGE CHANGES

increases is greater than the corresponding magnitude for the gross flow that decreases. In addition, the rise in job destruction when the wage increases is almost identical to the rise in job creation when the wage decreases. A similar observation holds for falls in job creation and rises in job destruction. Consequently, the responses of net job growth (not shown) are essentially symmetric across the two wage changes.

The right column of Figure 5 indicates that the adjustment costs dampen the responses of both job creation and destruction in the period of the wage changes. As in the frictionless case, the magnitude of the initial change in the gross job flow that increases is greater than the corresponding magnitude for the gross flow that decreases. In contrast to the two-period-cycle example, the adjustment costs appear to dampen creation and destruction by similar amounts in the period of the wage changes. The reason for this is that the policy asymmetry induced by

positive adjustment costs is much less pronounced here. In the current example, when the wage changes, the intercept of the destruction schedule changes by 0.64, while the intercept of the creation schedule changes by 0.50. That is, the destruction schedule shifts by only 28 percent more than the creation schedule, compared to more than twice as much in the two-period-cycle case. This effect of persistence in the wage is analogous to the effect of increasing the persistence of plants' idiosyncratic productivity in the example of Section I. When the wage change is permanent, the expected savings of creation costs tomorrow from creating a job today loom larger in the calculation of the cost of job creation, and the expected cost of reversing the creation decision is less important. Thus, the total expected discounted adjustment costs associated with the last job created fall. Similarly, the total expected discounted adjustment costs associated with the last job retained also decrease. This mitigates the asymmetric fluctu-

ations in the job creation and destruction schedules. Combined with the offsetting effects of the growth-rate asymmetry, the net impact of wage changes on creation and destruction is essentially symmetric when the wage changes are permanent.

A final observation worth making about Figure 5 is that positive adjustment costs introduce persistence into the responses of gross job flows. The key here is the region of inaction in the state space of the employment policy induced by proportional adjustment costs. As is well known, this feature of the model provides for an endogenous source of persistence in gross job flows since individual plants are slow to respond to wage changes.

C. Quantifying the Effects of Adjustment Costs

We now study the statistical properties of the model and compare them to deseasonalized quarterly data on gross job flows for the transportation equipment industry over the sample period 1972:II to 1988:IV.¹³ For this exercise we focus on the baseline calibration with positive adjustment costs (see Table 1). Note that the persistence of the wage in the baseline calibration is intermediate to the two cases considered above. In Table 2 we report statistics on gross job flows for the transportation equipment industry and statistics based on simulating the calibrated model.¹⁴ In addition to our baseline case we also consider the calibration based on zero adjustment costs. Other entries in the table correspond to experiments designed to illustrate various model properties. We adopt the notation used by Davis et al. (1996) to denote the components of gross job flows: job destruction is denoted by *NEG*, job creation by *POS*, job reallocation is denoted by *SUM*, where $SUM = NEG + POS$, and job growth is denoted by *NET*, where $NET = POS - NEG$.

Consider the calibrated frictionless case, $\tau_c = \tau_d = 0$, first. Notice that this model can

account for many features of the transportation equipment industry data. The calibration procedure guarantees that the standard deviation of *NET* in the model exactly matches that in the data. In the model and the data *POS* is strongly procyclical, *NEG* is strongly countercyclical, and *POS* and *NEG* are negatively correlated. These observations seem in line with the impressions given by Figures 4 and 5.

The frictionless model has two main shortcomings. First, it underpredicts the variance ratio, $v(NEG)/v(POS)$, the measure of asymmetries in gross job flows emphasized by Davis et al. (1996). Second, it predicts that job reallocation is acyclical; that is, the correlation of *SUM* and *NET* is essentially zero. The failure with respect to the variance ratio derives from underpredicting the volatility of *NEG* and overpredicting the volatility of *POS*. That *SUM* is acyclical is explained by the fact that the covariance between *NET* and *SUM* is equal to the variance of *POS* less the variance of *NEG*. Other weaknesses of the frictionless model are that *SUM* is too smooth and that *NET* is negatively autocorrelated. The former is due to the excessive negative correlation of *POS* and *NEG*, while the latter follows from the close association of job growth with wage growth (which is negatively autocorrelated here) when employment at the micro level is determined by a static labor-demand schedule.

Table 2 shows that the baseline model represents a substantial improvement over the frictionless model. Significantly, the variance ratio rises from roughly unity to 1.82. This is below the value in the data, but still substantial. The improvement in the variance ratio compared to the frictionless case is achieved by reducing the volatility of *POS* and amplifying fluctuations in *NEG*, in both cases moving the model closer to the data. These effects are due to asymmetric variation in the employment schedules: the destruction schedule shifts by 66 percent more than the creation schedule whenever the wage changes.

One way to gauge the importance of this asymmetry is to compare the baseline model to a case where we keep the parameters at their baseline values but artificially constrain the log creation and destruction schedules to shift by the same amount in response to a wage change. Specifically, we fix the schedules as follows:

¹³ The raw data were compiled by Davis et al. (1996). To deseasonalize this data we removed quarterly means.

¹⁴ All statistics for the model are based on simulating the model for 1,050 periods, starting at the steady-state distribution of employment across plants, and then discarding the first 50 observations.

TABLE 2—MODEL IMPLICATIONS FOR GROSS JOB FLOWS

Statistic ^a	Data ^b	Calibrated ^c		Perturbations of the baseline calibration ^d				
		Baseline	$\tau_c = 0$ $\tau_d = 0$	$\tau_c = 3$ $\tau_d = 3$	$\tau_c = 0.5$ $\tau_d = 0$	No trend	Manufacturing trend	$\rho = 0$
$v(NEG)/v(POS)$	2.99	1.82	1.02	3.83	1.18	1.75	1.97	2.37
$s(NEG)$	3.55	3.55	3.55	2.30	8.66	3.55	3.56	3.62
$s(SUM)$	2.73	0.76	0.60	1.80	2.20	0.74	0.81	0.76
$s(NEG)$	2.73	2.06	1.81	1.20	4.64	2.04	2.10	2.00
$s(POS)$	1.58	1.53	1.79	0.61	4.28	1.54	1.50	1.30
$c(POS, NEG)$	-0.30	-0.95	-0.95	-0.97	-0.88	-0.95	-0.95	-0.98
$c(NEG, SUM)$	-0.52	-0.71	-0.04	-0.95	-0.17	-0.69	-0.75	-0.93
$c(NEG, POS)$	0.67	0.98	0.99	0.99	0.97	0.99	0.98	0.99
$c(NEG, NEG)$	-0.91	-0.99	-0.99	-0.99	-0.97	-0.99	-0.99	-0.99
$\rho(NEG)$	0.08	0.08	-0.24	0.02	0.07	0.08	0.08	-0.34

^a $v(x)$ denotes the variance of variable x ; $s(x)$ denotes the standard deviation of variable x ; $c(x, y)$ denotes the correlation between variable x and variable y ; and $\rho(x)$ is the autocorrelation coefficient of variable x . Standard deviations are reported in percent. See the text for variable definitions.

^b Based on deseasonalized data for the transportation equipment industry from Davis et al. (1996).

^c Models calibrated to the transportation equipment industry as described in the text.

^d Models with all parameters identical to the baseline calibration except as indicated. In the “No trend” case $\mu_z = -0.0381$ so that $E(NEG) = 0$ and in the “Manufacturing trend” case $\mu_z = -0.0393$ so that $E(NEG) = -0.33$ percent.

Fixed creation schedule:

$$\ln \underline{n}_t = \ln \underline{y}(1) + \frac{1}{1 - \alpha} \ln z_t - \frac{1}{1 - \alpha} \ln W_t$$

Fixed destruction schedule:

$$\ln \bar{n}_t = \ln \bar{y}(1) + \frac{1}{1 - \alpha} \ln z_t - \frac{1}{1 - \alpha} \ln W_t$$

where $\underline{y}(1)$ and $\bar{y}(1)$ are the values of the scaled employment thresholds computed in a version of the model without aggregate uncertainty in which the wage is equal to unity. In this example there is no policy asymmetry and so gross job flows are expected to be dominated by the growth-rate asymmetry, which tends to make creation more volatile than destruction. Indeed, when we simulate this model the variance ratio falls below unity to 0.71.

Given the asymmetry in how *POS* and *NEG* vary, the model also predicts countercyclical job reallocation—the correlation of *NET* with *SUM* is strongly negative as it is in the data. Another improvement over the frictionless model can be seen in terms of the persistence of fluctuations in job growth. Despite the fact that wage changes are negatively autocorrelated, with positive adjustment costs it is possible to

exactly match the slightly positive autocorrelation of *NET* for the transportation equipment industry. Finally, incorporating adjustment costs does not improve the model's predictions with respect to the correlation between *POS* and *NEG*, which is still too strongly negative, and therefore *SUM* remains too smooth relative to the data. However, the fact that *POS* and *NEG* remain negatively correlated as they are in the data is a desirable feature of the model.

For the remainder of this section we discuss how various parameters influence the behavior of the model. In Table 2 we report results of two experiments that show the implications of assuming adjustment costs equal to the extreme values of the range described in Section III, subsection A. For these experiments all parameters are kept at their baseline values except for the adjustment costs. We consider the implications of assuming $\tau_c = \tau_d = 3$, and $\tau_c = 0.5$ and $\tau_d = 0$. The $\tau_c = \tau_d = 3$ case reveals that large adjustment costs give rise to large asymmetries in gross job flows. Except for the reduced volatility expected with larger adjustment costs, this version of the model has the same basic characteristics as the baseline. The $\tau_c = 0.5$ and $\tau_d = 0$ case implies a small variance ratio but one which is still significantly greater than unity and, volatility aside, the statistics conform with the baseline.

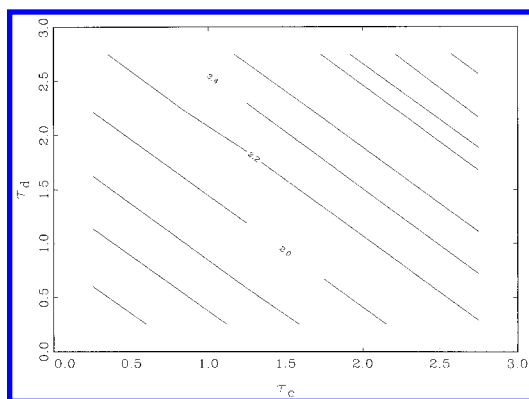


FIGURE 6. ISOQUANTS SHOWING THE IMPACT OF ADJUSTMENT COSTS ON THE DESTRUCTION-CREATION VARIANCE RATIO

One issue not addressed by these perturbations is the role of asymmetries in the adjustment costs themselves. We address this issue by considering the implications of values of τ_c and τ_d taken from a finely spaced grid of values within the ranges for these parameters discussed above. For each combination of τ_c and τ_d values, we recalibrate and simulate the model. Aside from their impact on the volatility of gross job flows, the main effect of alternative values of the adjustment costs is on the variance ratio statistic, and so we focus on this here. Figure 6 plots values of the adjustment costs which imply given values of the variance ratio. Moving northeast in the figure corresponds to values of the adjustment costs implying larger values of the variance ratio. The striking feature of the figure is that asymmetries in gross job flows appear to be invariant to the relative magnitude of the adjustment costs. Since the contour lines are approximately linear with slope equal to -1 the variance ratio depends only on the sum of the adjustment costs, $\tau_c + \tau_d$.

Given the results from Foote's (1998) experiments with trends in a (S, s) model, it is interesting to investigate how trends in employment growth influence our model's predictions. The "No trend" and "Manufacturing trend" experiments in Table 2 are designed for this purpose. Here we leave all parameters at their baseline values except for μ , which has been adjusted so that the model implies average employment growth equal to zero and -0.33 per-

cent per quarter, the value for the manufacturing sector as a whole over the sample period. As the table shows, trends in employment growth have only a modest effect on the variance ratio. Eliminating the trend from the baseline calibration lowers the variance ratio to 1.75, and lowering the trend in the baseline calibration by a factor of three increases the variance ratio to 1.97. The other statistics are not influenced very much either. While larger trends certainly have a greater impact on gross job flows, generally the effects of employment trends seem small relative to the influence of the policy asymmetry. Finally, notice that in the calibrated frictionless version of the model the variance ratio is slightly above unity. This does not reflect approximation or simulation error, but is due to the slight trend in industry employment and the forces discussed in footnote 5.

The last experiment in Table 2 illustrates the impact of autocorrelation in the wage. All parameters are again left at their baseline values, except for ρ , which is set to 0. Notice that the asymmetry in gross job flows increases to 2.37 in this case. This is consistent with the analysis in Section III, subsection B. Evidently, the persistence of wage changes is a crucial element in the model influencing the dynamics of gross job flows. As discussed before, this is due to the impact ρ has on the degree of policy asymmetry and the offsetting effects of the growth-rate asymmetry.

IV. Related Literature

This paper is related to two literatures that have addressed the evidence on gross job flows. The most closely related is work based on (S, s) models of employment dynamics. In these models, plants choose their actual employment to regulate the employment gap, the difference between actual employment and "desired employment." Two threshold points, L and U , and their associated return points, l and u , characterize the employment policy. A plant leaves its employment unchanged if the employment gap lies between L and U . If the employment gap equals or exceeds U in the absence of adjustment, the plant destroys jobs to bring the gap down to u , and if the employment gap would otherwise equal or fall below L , the plant creates jobs to bring it up to l . Bertola and

Caballero (1990) justify the optimality of this policy in the presence of fixed and proportional costs of adjustment with a particular return function. Common shocks to desired employment at all plants generate aggregate fluctuations, and idiosyncratic uncertainty ensures that both job creation and destruction occur in all periods.

The employment policy we study can be represented in this form, so there is a close connection between our results and results derived from (S, s) models. Define desired employment to be what a plant would choose in the absence of any adjustment costs. Then the employment gap, s_t , can be specified as the logarithmic difference between actual and desired employment:

$$(16) \quad s_t = \ln n_t - \frac{1}{1 - \alpha} (\ln z_t + \ln \alpha - \ln W_t).$$

Notice that W_t acts as an aggregate shock. For example, positive innovations to W_t reduce desired employment at all plants and shift the cross-sectional distribution of s_t to the right. Since there are no fixed costs in our model, the thresholds, U and L , equal their associated return points, u and l . The common threshold and return values are

$$U = u = \ln \bar{y}(W_t) - \frac{1}{1 - \alpha} (\ln \alpha - \ln W_t)$$

$$L = l = \ln \underline{y}(W_t) - \frac{1}{1 - \alpha} (\ln \alpha - \ln W_t).$$

In Caballero's (1992) (S, s) model the threshold and return points differ, are defined in levels, and do not vary. He explored the implications of assuming that the number of jobs destroyed by a contracting plant, $U - u$, is greater than the number of jobs created by an expanding plant, $l - L$. Focusing on the total number of jobs created and destroyed, rather than the job creation and destruction rates studied here, he showed that with a symmetrically distributed aggregate shock the microeconomic asymmetry is offset by dynamics in the numbers of plants creating and destroying jobs so that the total numbers of jobs created and destroyed are roughly equally variable. Caballero demon-

strated that the number of jobs destroyed could vary more than the number of jobs created if the aggregate shock is asymmetrically distributed in a particular way. The effects underlying this result are not present in our analysis because we assumed the aggregate shock, i.e., the wage, is symmetrically distributed.

Using a model similar to Caballero's, Foote (1998) introduced trends into plants' employment levels to break the tight inverse relationship between the intensity of job creation and destruction at a given plant and the numbers of plants creating and destroying jobs. A negative trend tends to increase the number of plants destroying jobs, increasing the variability of job destruction and reducing it for job creation. A positive trend has the opposite effect. Foote demonstrates that employment trends are unable to account quantitatively for the high observed variance ratios in the manufacturing sector. The small effects of trends in our model are consistent with these results.

The policy asymmetry and the growth-rate asymmetry are both absent from the standard (S, s) models studied by Caballero (1992) and Foote (1998), and this accounts for the differences between our results and theirs. Consider first the policy asymmetry. If the job creation and destruction schedules and the frictionless employment schedule [the schedule determining desired employment in (16)] varied symmetrically with aggregate shocks, so that $\ln \bar{y}(W_t)$ and $\ln \underline{y}(W_t)$ equaled constants minus $(1/(1 - \alpha)) \ln W_t$, then u and l would be time invariant, as in a standard (S, s) model. However, $\ln \underline{y}(W_t)$ moves less than $\ln \bar{y}(W_t)$ and both of these variables fluctuate less than the intercept of the frictionless schedule, $(\ln \alpha - \ln W_t)/(1 - \alpha)$. Therefore, u and l both covary positively with the wage and u fluctuates less than l . As in a standard (S, s) model, variation in the aggregate shock, W_t , shifts the distribution of s_t . Unlike in a standard model, it also shifts the return points in the same direction as the distribution of s_t , dampening the responses of both job creation and destruction. Because u moves less than l , this dampening affects job destruction less than job creation and generates the observed variance asymmetry.

As we noted above, the growth-rate asymmetry tends to offset the effects of the policy asymmetry. This was evident in the experi-

ment in Section III, subsection C, where we artificially eliminated the policy asymmetry by constraining the job creation and destruction schedules to fluctuate symmetrically, which corresponds to the case where the return points u and l are fixed over time. In that experiment, the job creation rate is more volatile than the job destruction rate. Since Caballero (1992) specifies his model in levels, not logarithms, and focuses on gross job flows in levels, not rates, there is no growth-rate asymmetry in his analysis. Foote (1998) uses the logarithmic approximation to the gross job flow rates and so his analysis also does not exhibit a growth-rate asymmetry. This is why these papers find that fixed return points without asymmetric aggregate shocks and employment trends imply symmetric gross job flows.

The second related literature focuses on frictions in the movement of workers across jobs. Prominent examples include Caballero and Hammour's (1994) model of creative destruction, Mortensen and Pissarides' (1994) job matching model, and Caballero and Hammour's (1996) synthesis and extension of these two models. To focus on reallocation frictions, this research typically assumes a simple production structure in which a plant or "production unit" consists of a Leontief technology combining a unit of capital and a unit of labor. Variation in gross job flows is synonymous with variation in the number of plants. This contrasts with the current analysis in which variation in gross job flows arises due to variation in the utilization of production capacity at individual plants.

When subjected to transitory aggregate productivity shocks, job matching models tend to concentrate both job creation and destruction in recessions, when the opportunity cost of moving resources out of production and into matching is temporarily low. Thus, these models tend to predict counterfactually that job creation and destruction co-move positively. To remedy this perceived shortcoming, Caballero and Hammour (1996) consider incomplete contracting between matched employer-employee pairs, which decouples creation and destruction so that they are not positively correlated. These frictions do not endogenously generate a relatively high volatility of job destruction. Caballero and Hammour accomplish this by adding

congestion effects in entry so that there is an incentive to smooth creation over time. Neither of these frictions is present in our model.

This second literature abstracts from our focus, namely the implications for gross job flows of the technological determinants of employment at a fixed number of plants. Likewise, this paper and most of the literature using (S, s) models of employment abstracts from the costs of moving workers across plants. We view the latter abstraction as a sensible starting point for taking into account the fact documented by Davis et al. (1996), that by far most of the job creation and destruction in the manufacturing sector is due to employment variation at individual incumbent plants rather than at entering and exiting plants.¹⁵ Nevertheless, it seems clear that enhancing our model with a richer description of how workers move across plants would be fruitful. Job creation and destruction co-move strongly negatively in our model and in (S, s) models because aggregate shocks alter the desired employment of all plants in the same direction. Introducing matching frictions that make hiring less expensive for plants during times of high unemployment could induce a source of positive co-movement between job creation and destruction that our model lacks.

V. Conclusion

We have described a model in which asymmetric variation in gross job flows arises naturally from profit maximization in the presence of proportional job creation or destruction costs. The adjustment costs cause employment at shrinking plants to respond by more to aggregate shocks to the cost of a job than employment at growing plants. Our quantitative analysis confirmed that this mechanism can account for asymmetric variation in job creation and destruction of the kind observed in the U.S. manufacturing sector. In addition, we found the mechanism to have the strongest influence on gross job flows when disturbances to the cost of a job are expected to

¹⁵ For example, over the period 1972:II to 1988:IV, only 8.4 percent of total job creation is accounted for by startup plants, while just 11.6 percent of total job destruction is accounted for by plant shutdowns. See Davis et al. (1996 Figure 2.3, p. 29).

be short-lived. Interpreting the cost shocks broadly as being due to changes in both supply and demand, along the lines suggested by our general-equilibrium model, these findings suggest a new way of understanding the evidence on gross job flows.

Although our model has a general-equilibrium interpretation, the assumptions of constant marginal utility of consumption, infinitely elastic labor supply, and no saving are too restrictive for quantitative general-equilibrium business-cycle studies. Accordingly, it would be useful to incorporate our model of plant dynamics into a standard dynamic general-equilibrium business-cycle framework. Doing so may improve the predictions of the standard model for gross job flows, but may be beneficial for another reason as well. A notorious shortcoming of these models is their absence of an endogenous propagation mechanism. The adjustment costs we have considered add a natural source of propagation due to the time it takes for individual plants to respond to shocks.

As we mentioned above, including a more realistic description of worker flows between jobs in our model is likely to improve its description of the data. Recent empirical work by Caballero et al. (1997) suggests another useful extension of our model. This work provides support for the view that nonconvexities, such as those which arise due to fixed adjustment costs, are important for explaining employment variation at manufacturing plants. It would be interesting to determine whether the microeconomic asymmetries we study here extend to environments with nonconvexities. We conjecture that a general two-sided (S , s) rule with time-varying threshold and return points would characterize plants' optimal employment policies under fixed adjustment costs and that first-order conditions similar to (9) and (10) would characterize the optimal return points. In this case we expect a policy asymmetry to characterize the return points, which would contribute to asymmetries in gross job flows. Verifying this conjecture is left to future research.

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